

Code-Layout, Readability and Reusability

Date

06 July 2026

Team ID

Project Name

Smart Lender — Applicant Credibility Prediction for Loan Approval

Maximum Marks

5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space PEP8 indentation throughout app.py and the notebook
2	Proper File Structure	Files and folders are logically organized	Yes	Separated into templates/, static/css/, notebook, and model artifacts
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. x_bal, y_bal, X_train, scale, model
4	Function / Method Names	Functions are descriptively named	Yes	decisionTree(), randomForest(), knn(), xgb(), evaluate()
5	Code Comments	Inline and block comments explain logic	Yes	Each notebook section and key app.py block is commented
6	Modular Design	Code is split into reusable functions/modules	Yes	Each model has its own training/evaluation function; Flask routes are separated by responsibility
7	No Redundant Code	Duplicate or unused code is removed	Yes	Removed an earlier fragile pickle-based model save in favor of one consistent native-format save
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Form fields are validated as required on the frontend; backend assumes well-formed numeric input

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Reusability Level	Where Reused
1	Model evaluation function (evaluate())	Python	High	Reused identically across all 4 model trainings in the notebook
2	style.css design tokens	CSS	High	Shared across home.html, input.html, and output.html
3	Flask route structure (home/predict/submit)	Python	Medium	Pattern can be reused for any future additional form-based ML feature

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of notebook, backend, templates, and static assets
Readability	4	Descriptive names and comments throughout; could add more docstrings
Reusability	4	Core training/evaluation logic is reusable across all four models
Documentation / Comments	4	README and RESULTS files document setup and outcomes
Overall Score	4.25	Solid, maintainable codebase for the project's scope